

Gait & Posture

Evaluation of IMU-based foot-strike detection algorithms for outdoor walking across surfaces

--Manuscript Draft--

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Order of Authors:	Jill Emmerzaal, PhD Philippe C Dixon, PhD
Abstract:	Background: Accurate event detection is crucial in gait analysis, as small timing errors can affect biomechanical outcomes. This study investigates the performance of inertial measurement unit (IMU)-based foot-strike detection algorithms for outdoor gait analysis across multiple walking surfaces. Methods: Using a publicly available dataset of 20 participants walking in an urban environment, we compared 16 open-access algorithms based on shank and lower-back IMUs against pressure insole ground truth. Algorithms were assessed for detection accuracy and timing error, on level ground, slopes (up and down), stairs (up and down), and curbs (stepping up and stepping down). Results: Results show that the Greene-McGrath algorithm outperformed all other algorithms, achieving the highest overall accuracy (98.5\% $\pm$ 0.4\%) and lowest mean timing error (9.37ms $\pm$ 6.84ms). Performance remained robust across most surfaces, stair ascend showed reduced accuracy (88\%) and increased variability (5.2\%). Significance: We recommend using the Greene-McGrath algorithm for outdoor data collection as the most reliable choice. Future work is needed to improve foot-strike detection on stairs and evaluate performance in populations with gait impairments.

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Editor-in-Chief
Gait & Posture
Elsevier
Amsterdam, The Netherlands

Dear Editor,

We are pleased to submit our manuscript entitled “**Evaluation of IMU-based foot-strike detection algorithms for outdoor walking across surfaces**” for consideration for publication in *Gait & Posture* as a short communication.

Within laboratory settings, methods to detect gait events have been developed that rely on either force plate or marker data. IMU-based step detection methods have been developed as an alternative. However, it is unclear if performance is affected by outdoor walking surface.

In this study, we assessed the accuracy and timing error of IMU-based foot-strike detection algorithms for outdoor gait analysis across common surfaces. For this we used 16 pre-existing algorithms. Our results demonstrate that for the use case of event detection during outdoor walking, we recommend using the Greene & McGrath method; nonetheless, users should be cautious when interpreting results on stairs. We believe this represents a significant contribution to the field of outdoor biomechanical gait analysis.

We believe this manuscript is well-suited for *Gait & Posture* as a short communication, given its alignment with the journal’s focus on “Techniques for the measurement of gait and posture”. The short communication format is particularly appropriate, as this work introduces a novel application of established techniques rather than a full methodological contribution.

This manuscript has not been published previously and is not under consideration for publication elsewhere. All authors have approved the manuscript and agree with its submission to *Gait & Posture*.

Sincerely,

Philippe C. Dixon, PhD
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Highlights

- Sixteen shank and lower back IMU-based foot-strike detection algorithms were tested across multiple outdoor walking surfaces.
- The Greene & McGrath algorithm achieved the highest accuracy (98.5%) and lowest timing error (9.37 ms) overall.
- Algorithm performance remained robust across level ground, slopes, curbs and stair descend but degraded on stair ascend (88% accuracy)
- The Greene & McGrath algorithm is recommended for outdoor gait analysis, pending pending modifications for stair ascent and further validation in populations with gait impairments.

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33 LaTeX Font Info: Overwriting symbol font `largesymbols' in version
34 `bold'
35 (Font) OMX/txex/m/n --> OMX/txex/bx/n on input line 121.
36 \symlargesymbolsA=\mathgroup9
37 LaTeX Font Info: Overwriting symbol font `largesymbolsA' in version
38 `bold'
39 (Font) U/txexa/m/n --> U/txexa/bx/n on input line 129.
40 LaTeX Font Info: Redefining math symbol \mathsterling on input line
41 164.
42 LaTeX Font Info: Redefining math symbol \hbar on input line 591.
43 LaTeX Info: Redefining \not on input line 1043.
44 ) (./ecrc.sty
45 Package: ecrc 2011/05/07 v1.2a ecrc
46 \dummylogowidth=\dimen157
47 ) (c:/texlive/2024/texmf-dist/tex/latex/caption/caption.sty
48 Package: caption 2023/08/05 v3.6o Customizing captions (AR)
49 (c:/texlive/2024/texmf-dist/tex/latex/caption/caption3.sty
50 Package: caption3 2023/07/31 v2.4d caption3 kernel (AR)
51 \caption@tempdima=\dimen158
52 \captionmargin=\dimen159
53 \caption@leftmargin=\dimen160
54 \caption@rightmargin=\dimen161
55 \caption@width=\dimen162
56 \caption@indent=\dimen163
57 \caption@parindent=\dimen164
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(c:/texlive/2024/texmf-dist/tex/latex/caption/caption-elsarticle.sto
File: caption-elsarticle.sto 2020/08/22 v2.0 Adaption of the caption
package to
the elsarticle document class (AR)
))
\c@caption@flags=\count280
\c@continuedfloat=\count281
) (c:/texlive/2024/texmf-dist/tex/latex/caption/subcaption.sty
Package: subcaption 2023/07/28 v1.6b Sub-captions (AR)
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Package caption Info: New subtype `subtable' on input line 238.
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) (c:/texlive/2024/texmf-dist/tex/latex/amsfonts/amssymb.sty
Package: amssymb 2013/01/14 v3.01 AMS font symbols
(c:/texlive/2024/texmf-dist/tex/latex/amsfonts/amsfonts.sty
Package: amsfonts 2013/01/14 v3.01 Basic AMSFonts support
\@emptytoks=\toks21
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(Font) U/txsya/m/n --> U/msa/m/n on input line 59.
LaTeX Font Info: Overwriting symbol font `AMSA' in version `bold'
(Font) U/txsya/bx/n --> U/msa/m/n on input line 59.
LaTeX Font Info: Redefining symbol font `AMSb' on input line 60.
LaTeX Font Info: Overwriting symbol font `AMSb' in version `normal'
(Font) U/txsyb/m/n --> U/msb/m/n on input line 60.
LaTeX Font Info: Overwriting symbol font `AMSb' in version `bold'
(Font) U/txsyb/bx/n --> U/msb/m/n on input line 60.
LaTeX Font Info: Redefining math delimiter \ulcorner on input line
74.
LaTeX Font Info: Redefining math delimiter \urcorner on input line
75.
LaTeX Font Info: Redefining math delimiter \llcorner on input line
76.
LaTeX Font Info: Redefining math delimiter \lrcorner on input line
77.
LaTeX Font Info: Redefining math symbol \square on input line 141.
LaTeX Font Info: Redefining math symbol \lozenge on input line 142.
)
LaTeX Font Info: Redefining math symbol \boxdot on input line 44.
LaTeX Font Info: Redefining math symbol \boxplus on input line 45.
LaTeX Font Info: Redefining math symbol \boxtimes on input line 46.
LaTeX Font Info: Redefining math symbol \blacksquare on input line
48.
LaTeX Font Info: Redefining math symbol \centerdot on input line 49.
LaTeX Font Info: Redefining math symbol \blacklozenge on input line
51.
LaTeX Font Info: Redefining math symbol \circlearrowright on input
line 52.
LaTeX Font Info: Redefining math symbol \circlearrowleft on input
line 53.

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LaTeX Font Info: Redefining math symbol `\leftrightharpoons` on input line 56.
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 LaTeX Font Info: Redefining math symbol `\leftleftarrows` on input line 63.
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LaTeX Font Info: Redefining math symbol $\doteqdot$ on input line 89.
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 LaTeX Font Info: Redefining math symbol $\risingdotseq$ on input line 104.
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LaTeX Font Info: Redefining math symbol $\Lleftarrow$ on input line 131.
 LaTeX Font Info: Redefining math symbol $\veebar$ on input line 132.
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 LaTeX Font Info: Redefining math symbol $\Bumpeq$ on input line 155.
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LaTeX Font Info: Redefining math symbol \nsucc on input line 178.
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 LaTeX Font Info: Redefining math symbol \lneq on input line 183.
 LaTeX Font Info: Redefining math symbol \gneq on input line 184.
 LaTeX Font Info: Redefining math symbol \npreceq on input line 185.
 LaTeX Font Info: Redefining math symbol \nsucceq on input line 186.
 LaTeX Font Info: Redefining math symbol \precnsim on input line 187.
 LaTeX Font Info: Redefining math symbol \succnsim on input line 188.
 LaTeX Font Info: Redefining math symbol \lnsim on input line 189.
 LaTeX Font Info: Redefining math symbol \gnsim on input line 190.
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 195.
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 LaTeX Font Info: Redefining math symbol `\succapprox` on input line 255.

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4 LaTeX Font Info: Redefining math symbol \precapprox on input line
5 256.
6 LaTeX Font Info: Redefining math symbol \curvearrowleft on input line
7 257.
8 LaTeX Font Info: Redefining math symbol \curvearrowright on input
9 line 258.
10
11 LaTeX Font Info: Redefining math symbol \digamma on input line 259.
12 LaTeX Font Info: Redefining math symbol \varkappa on input line 260.
13 LaTeX Font Info: Redefining math symbol \Bbbk on input line 261.
14 LaTeX Font Info: Redefining math symbol \hslash on input line 262.
15 LaTeX Font Info: Redefining math symbol \backepsilon on input line
16 265.
17
18 ) (c:/texlive/2024/texmf-dist/tex/latex/hyperref/hyperref.sty
19 Package: hyperref 2024-07-10 v7.01j Hypertext links for LaTeX
20 (c:/texlive/2024/texmf-dist/tex/latex/kvsetkeys/kvsetkeys.sty
21 Package: kvsetkeys 2022-10-05 v1.19 Key value parser (HO)
22 ) (c:/texlive/2024/texmf-dist/tex/generic/kvdefinekeys/kvdefinekeys.sty
23 Package: kvdefinekeys 2019-12-19 v1.6 Define keys (HO)
24 ) (c:/texlive/2024/texmf-dist/tex/generic/pdfescape/pdfescape.sty
25 Package: pdfescape 2019/12/09 v1.15 Implements pdfTeX's escape features
26 (HO)
27 (c:/texlive/2024/texmf-dist/tex/generic/ltxcmds/ltxcmds.sty
28 Package: ltxcmds 2023-12-04 v1.26 LaTeX kernel commands for general use
29 (HO)
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31 Package: pdftextcmds 2020-06-27 v0.33 Utility functions of pdfTeX for
32 LuaTeX (HO)
33 )
34 (c:/texlive/2024/texmf-dist/tex/generic/infwarerr/infwarerr.sty
35 Package: infwarerr 2019/12/03 v1.5 Providing info/warning/error messages
36 (HO)
37 )
38 )
39 Package pdftextcmds Info: \pdf@primitive is available.
40 Package pdftextcmds Info: \pdf@ifprimitive is available.
41 Package pdftextcmds Info: \pdfdraftmode found.
42 )) (c:/texlive/2024/texmf-dist/tex/latex/hycolor/hycolor.sty
43 Package: hycolor 2020-01-27 v1.10 Color options for hyperref/bookmark
44 (HO)
45 ) (c:/texlive/2024/texmf-dist/tex/latex/hyperref/nameref.sty
46 Package: nameref 2023-11-26 v2.56 Cross-referencing by name of section
47 (c:/texlive/2024/texmf-dist/tex/latex/refcount/refcount.sty
48 Package: refcount 2019/12/15 v3.6 Data extraction from label references
49 (HO)
50 ) (c:/texlive/2024/texmf-
51 dist/tex/generic/gettitlestring/gettitlestring.sty
52 Package: gettitlestring 2019/12/15 v1.6 Cleanup title references (HO)
53 (c:/texlive/2024/texmf-dist/tex/latex/kvoptions/kvoptions.sty
54 Package: kvoptions 2022-06-15 v3.15 Key value format for package options
55 (HO)
56 ))
57 \c@section@level=\count284
58 ) (c:/texlive/2024/texmf-dist/tex/generic/stringenc/stringenc.sty
59
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Package: stringenc 2019/11/29 v1.12 Convert strings between diff.
encodings (HO
)
)
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\Hy@linkcounter=\count285
\Hy@pagecounter=\count286
(c:/texlive/2024/texmf-dist/tex/latex/hyperref/pdhlenc.def
File: pdhlenc.def 2024-07-10 v7.01j Hyperref: PDFDocEncoding definition
(HO)
Now handling font encoding PD1 ...
... no UTF-8 mapping file for font encoding PD1
) (c:/texlive/2024/texmf-dist/tex/generic/intcalc/intcalc.sty
Package: intcalc 2019/12/15 v1.3 Expandable calculations with integers
(HO)
)
\Hy@SavedSpaceFactor=\count287
(c:/texlive/2024/texmf-dist/tex/latex/hyperref/puenc.def
File: puenc.def 2024-07-10 v7.01j Hyperref: PDF Unicode definition (HO)
Now handling font encoding PU ...
... no UTF-8 mapping file for font encoding PU
)
Package hyperref Info: Hyper figures OFF on input line 4157.
Package hyperref Info: Link nesting OFF on input line 4162.
Package hyperref Info: Hyper index ON on input line 4165.
Package hyperref Info: Plain pages OFF on input line 4172.
Package hyperref Info: Backreferencing OFF on input line 4177.
Package hyperref Info: Implicit mode ON; LaTeX internals redefined.
Package hyperref Info: Bookmarks ON on input line 4424.
\c@Hy@tempcnt=\count288
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Package: url 2013/09/16 ver 3.4 Verb mode for urls, etc.
)
LaTeX Info: Redefining \url on input line 4763.
\XeTeXLinkMargin=\dimen167
(c:/texlive/2024/texmf-dist/tex/generic/bitset/bitset.sty
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(c:/texlive/2024/texmf-dist/tex/generic/bigintcalc/bigintcalc.sty
Package: bigintcalc 2019/12/15 v1.5 Expandable calculations on big
integers (HO
)
))
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\Field@Width=\dimen168
\Fld@charsize=\dimen169
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Package hyperref Info: Hyper index ON on input line 6050.
Package hyperref Info: backreferencing OFF on input line 6057.
Package hyperref Info: Link coloring OFF on input line 6062.
Package hyperref Info: Link coloring with OCG OFF on input line 6067.
Package hyperref Info: PDF/A mode OFF on input line 6072.
(c:/texlive/2024/texmf-dist/tex/latex/base/atbegshi-ltx.sty

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4 Package: atbegshi-ltx 2021/01/10 v1.0c Emulation of the original atbegshi
5 package with kernel methods
6 )
7 \Hy@abspage=\count290
8 \c@Item=\count291
9 \c@Hfootnote=\count292
10 )
11 Package hyperref Info: Driver (autodetected): hpdftex.
12 (c:/texlive/2024/texmf-dist/tex/latex/hyperref/hpdftex.def
13 File: hpdftex.def 2024-07-10 v7.01j Hyperref driver for pdfTeX
14 (c:/texlive/2024/texmf-dist/tex/latex/base/atveryend-ltx.sty
15 Package: atveryend-ltx 2020/08/19 v1.0a Emulation of the original
16 atveryend pac
17 kage
18 with kernel methods
19 )
20 \Fld@listcount=\count293
21 \c@bookmark@seq@number=\count294
22 (c:/texlive/2024/texmf-dist/tex/latex/rerunfilecheck/rerunfilecheck.sty
23 Package: rerunfilecheck 2022-07-10 v1.10 Rerun checks for auxiliary files
24 (HO)
25 (c:/texlive/2024/texmf-dist/tex/generic/uniquecounter/uniquecounter.sty
26 Package: uniquecounter 2019/12/15 v1.4 Provide unlimited unique counter
27 (HO)
28 )
29 Package uniquecounter Info: New unique counter `rerunfilecheck' on input
30 line 2
31 85.
32 )
33 \Hy@SectionHShift=\skip55
34 ) (c:/texlive/2024/texmf-dist/tex/latex/booktabs/booktabs.sty
35 Package: booktabs 2020/01/12 v1.61803398 Publication quality tables
36 \heavyrulewidth=\dimen170
37 \lightrulewidth=\dimen171
38 \cmidrulewidth=\dimen172
39 \belowrulesep=\dimen173
40 \belowbottomsep=\dimen174
41 \aboverulesep=\dimen175
42 \abovetopsep=\dimen176
43 \cmidrulesep=\dimen177
44 \cmidrulekern=\dimen178
45 \defaultaddspace=\dimen179
46 \@cmidla=\count295
47 \@cmidlb=\count296
48 \@aboverulesep=\dimen180
49 \@belowrulesep=\dimen181
50 \@thisruleclass=\count297
51 \@lastruleclass=\count298
52 \@thisrulewidth=\dimen182
53 ) (c:/texlive/2024/texmf-dist/tex/latex/gensymb/gensymb.sty
54 Package: gensymb 2022/10/17 v1.0.2 (KJH)
55 ) (c:/texlive/2024/texmf-dist/tex/latex/censor/censor.sty
56 Package: censor 2023/06/05\ 4.3\ Provides capability for redaction of
57 sensitive
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information
(c:/texlive/2024/texmf-dist/tex/latex/pbox/pbox.sty
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(c:/texlive/2024/texmf-dist/tex/latex/tools/calc.sty
Package: calc 2023/07/08 v4.3 Infix arithmetic (KKT,FJ)
\calc@Acount=\count299
\calc@Bcount=\count300
\calc@Adimen=\dimen183
\calc@Bdimen=\dimen184
\calc@Askip=\skip56
\calc@Bskip=\skip57
LaTeX Info: Redefining \setlength on input line 80.
LaTeX Info: Redefining \addtolength on input line 81.
\calc@Ccount=\count301
\calc@Cskip=\skip58
) (c:/texlive/2024/texmf-dist/tex/latex/base/ifthen.sty
Package: ifthen 2024/03/16 v1.1e Standard LaTeX ifthen package (DPC)
)
\pb@xlen=\skip59
)
\censorruledepth=\skip60
\censorruleheight=\skip61
(c:/texlive/2024/texmf-dist/tex/generic/tokcycle/tokcycle.sty
(c:/texlive/2024/
texmf-dist/tex/generic/tokcycle/tokcycle.tex
\tc@tok=\toks22
\tcdepth=\count302
\cytoks=\toks23
)
Package: tokcycle 2021/08/25 V1.42 Cycle through and transform a stream
of token
ns
)) (c:/texlive/2024/texmf-dist/tex/latex/graphics/rotating.sty
Package: rotating 2016/08/11 v2.16d rotated objects in LaTeX
\c@r@tfl@t=\count303
\rotFPtop=\skip62
\rotFPbot=\skip63
\rot@float@box=\box59
\rot@mess@toks=\toks24
)
LaTeX Font Info: Trying to load font information for Tl+txr on input
line 25
.
(c:/texlive/2024/texmf-dist/tex/latex/txfonts/tltxr.fd
File: tltxr.fd 2000/12/15 v3.1
) (./main.aux)
\openout1 = `main.aux'.

LaTeX Font Info: Checking defaults for OML/txmi/m/it on input line 25.
LaTeX Font Info: Trying to load font information for OML+txmi on input
line
25.
(c:/texlive/2024/texmf-dist/tex/latex/txfonts/omltxmi.fd
File: omltxmi.fd 2000/12/15 v3.1

```

```

)
LaTeX Font Info:    ... okay on input line 25.
LaTeX Font Info:    Checking defaults for OMS/txsy/m/n on input line 25.
LaTeX Font Info:    Trying to load font information for OMS+txsy on input
line
25.
(c:/texlive/2024/texmf-dist/tex/latex/txfonts/omstxsy.fd
File: omstxsy.fd 2000/12/15 v3.1
)
LaTeX Font Info:    ... okay on input line 25.
LaTeX Font Info:    Checking defaults for OT1/cmr/m/n on input line 25.
LaTeX Font Info:    ... okay on input line 25.
LaTeX Font Info:    Checking defaults for T1/cmr/m/n on input line 25.
LaTeX Font Info:    ... okay on input line 25.
LaTeX Font Info:    Checking defaults for TS1/cmr/m/n on input line 25.
LaTeX Font Info:    ... okay on input line 25.
LaTeX Font Info:    Checking defaults for OMX/txex/m/n on input line 25.
LaTeX Font Info:    Trying to load font information for OMX+txex on input
line
25.
(c:/texlive/2024/texmf-dist/tex/latex/txfonts/omxtxex.fd
File: omxtxex.fd 2000/12/15 v3.1
)
LaTeX Font Info:    ... okay on input line 25.
LaTeX Font Info:    Checking defaults for U/txexa/m/n on input line 25.
LaTeX Font Info:    Trying to load font information for U+txexa on input
line 2
5.
(c:/texlive/2024/texmf-dist/tex/latex/txfonts/utxexa.fd
File: utxexa.fd 2000/12/15 v3.1
)
LaTeX Font Info:    ... okay on input line 25.
LaTeX Font Info:    Checking defaults for PD1/pdf/m/n on input line 25.
LaTeX Font Info:    ... okay on input line 25.
LaTeX Font Info:    Checking defaults for PU/pdf/m/n on input line 25.
LaTeX Font Info:    ... okay on input line 25.
(c:/texlive/2024/texmf-dist/tex/context/base/mkii/supp-pdf.mkii
[Loading MPS to PDF converter (version 2006.09.02).]
\scratchcounter=\count304
\scratchdimen=\dimen185
\scratchbox=\box60
\nofMPsegments=\count305
\nofMParguments=\count306
\everyMPshowfont=\toks25
\MPscratchCnt=\count307
\MPscratchDim=\dimen186
\MPnumerator=\count308
\makeMPintoPDFobject=\count309
\everyMPtoPDFconversion=\toks26
) (c:/texlive/2024/texmf-dist/tex/latex/epstopdf-pkg/epstopdf-base.sty
Package: epstopdf-base 2020-01-24 v2.11 Base part for package epstopdf
(c:/texlive/2024/texmf-dist/tex/latex/grfext/grfext.sty
Package: grfext 2019/12/03 v1.3 Manage graphics extensions (HO)
)

```

```

1
2
3
4 Package epstopdf-base Info: Redefining graphics rule for '.eps' on input
5 line 4
6 85.
7 Package grfext Info: Graphics extension search list:
8 (grfext)
9 [.pdf,.png,.jpg,.mps,.jpeg,.jbig2,.jb2,.PDF,.PNG,.JPG,.JPE
10 G,.JBIG2,.JB2,.eps]
11 (grfext) \AppendGraphicsExtensions on input line 504.
12 (c:/texlive/2024/texmf-dist/tex/latex/latexconfig/epstopdf-sys.cfg
13 File: epstopdf-sys.cfg 2010/07/13 v1.3 Configuration of (r)epstopdf for
14 TeX Liv
15 e
16 ))
17 *geometry* driver: auto-detecting
18 *geometry* detected driver: pdftex
19 *geometry* verbose mode - [ preamble ] result:
20 * driver: pdftex
21 * paper: custom
22 * layout: <same size as paper>
23 * layoutoffset:(h,v)=(0.0pt,0.0pt)
24 * hratio: 1:1
25 * vratio: 1:1
26 * modes:
27 * h-part:(L,W,R)=(64.75394pt, 468.0pt, 64.75394pt)
28 * v-part:(T,H,B)=(111.52342pt, 622.0pt, 111.52342pt)
29 * \paperwidth=597.50787pt
30 * \paperheight=845.04684pt
31 * \textwidth=468.0pt
32 * \textheight=622.0pt
33 * \oddsidemargin=-7.51605pt
34 * \evensidemargin=-7.51605pt
35 * \topmargin=17.25343pt
36 * \headheight=10.0pt
37 * \headsep=12.0pt
38 * \topskip=10.0pt
39 * \footskip=12.0pt
40 * \marginparwidth=65.0pt
41 * \marginparsep=11.0pt
42 * \columnsep=24.0pt
43 * \skip\footins=24.0pt plus 2.0pt minus 12.0pt
44 * \hoffset=0.0pt
45 * \voffset=0.0pt
46 * \mag=1000
47 * \@twocolumnfalse
48 * \@twosidefalse
49 * \@mparswitchfalse
50 * \@reversemarginfalse
51 * (lin=72.27pt=25.4mm, 1cm=28.453pt)
52
53 Package caption Info: Begin \AtBeginDocument code.
54 Package caption Info: hyperref package is loaded.
55 Package caption Info: rotating package is loaded.
56 Package caption Info: End \AtBeginDocument code.
57 Package hyperref Info: Link coloring OFF on input line 25.
58
59
60
61
62
63
64
65

```



```
1
2
3
4      (./main.out) (./main.out)
5      \@outlinefile=\write4
6      \openout4 = `main.out'.
7
```

```
8      LaTeX Info: Redefining \celsius on input line 25.
9      Package gensymb Info: Faking symbols for \degree and \celsius on input
10     line 25.
11
```

```
12
13
14     Package gensymb Warning: Not defining \perthousand.
15
```

```
16     LaTeX Info: Redefining \ohm on input line 25.
17     Package gensymb Info: Using \Omega for \ohm on input line 25.
18
```

```
19     Package gensymb Warning: Not defining \micro.
20
```

```
21     LaTeX Font Info:      Font shape `T1/txr/b/n' in size <10> not available
22     (Font)                Font shape `T1/txr/bx/n' tried instead on input line
23     37.
24
```

```
25     LaTeX Font Info:      Font shape `T1/txr/b/n' in size <9> not available
26     (Font)                Font shape `T1/txr/bx/n' tried instead on input line
27     38.
28
```

```
29     LaTeX Font Info:      Trying to load font information for OT1+txr on input
30     line 4
31     2.
32
```

```
33     (c:/texlive/2024/texmf-dist/tex/latex/txfonts/otltxr.fd
34     File: otltxr.fd 2000/12/15 v3.1
35     )
36
```

```
37     LaTeX Font Info:      Trying to load font information for U+txmia on input
38     line 4
39     2.
40
```

```
41     (c:/texlive/2024/texmf-dist/tex/latex/txfonts/utxmia.fd
42     File: utxmia.fd 2000/12/15 v3.1
43     )
44
```

```
45     LaTeX Font Info:      Trying to load font information for U+msa on input
46     line 42.
47
```

```
48     (c:/texlive/2024/texmf-dist/tex/latex/amsfonts/umsa.fd
49     File: umsa.fd 2013/01/14 v3.01 AMS symbols A
50     )
51
```

```
52     LaTeX Font Info:      Trying to load font information for U+msb on input
53     line 42.
54
```

```
55     (c:/texlive/2024/texmf-dist/tex/latex/amsfonts/umsb.fd
56     File: umsb.fd 2013/01/14 v3.01 AMS symbols B
57     )
58
```

```
59     LaTeX Font Info:      Trying to load font information for U+txsyc on input
60     line 4
61     2.
62
```

```
63     (c:/texlive/2024/texmf-dist/tex/latex/txfonts/utxsyc.fd
64     File: utxsyc.fd 2000/12/15 v3.1
65     )
66
```

```
67     LaTeX Font Info:      Trying to load font information for TS1+txr on input
68     line 4
69
```

5.
(c:/texlive/2024/texmf-dist/tex/latex/txfonts/tsltxr.fd
File: tsltxr.fd 2000/12/15 v3.1
)
LaTeX Font Info: Calculating math sizes for size <13> on input line
51.
LaTeX Font Info: Calculating math sizes for size <9.09996> on input
line 51.

Package natbib Warning: Citation `dumphart2025effect' on page 1 undefined
on in
put line 55.

Package natbib Warning: Citation `mickelborough2000validity' on page 1
undefine
d on input line 57.

Package natbib Warning: Citation `kiernan2023unsupervised' on page 1
undefined
on input line 57.

Package natbib Warning: Citation `panebianco2018analysis' on page 1
undefined o
n input line 57.

! LaTeX Error: File `elsevier-logo-3p' not found.

See the LaTeX manual or LaTeX Companion for explanation.
Type H <return> for immediate help.
...

1.61

I could not locate the file with any of these extensions:
.pdf,.png,.jpg,.mps,.jpeg,.jbig2,.jb2,.PDF,.PNG,.JPG,.JPEG,.JBIG2,.JB2,.e
ps
Try typing <return> to proceed.
If that doesn't work, type X <return> to quit.

! LaTeX Error: File `SDlogo-3p' not found.

See the LaTeX manual or LaTeX Companion for explanation.
Type H <return> for immediate help.
...

1.61

1
2
3
4
5 I could not locate the file with any of these extensions:
6 .pdf,.png,.jpg,.mps,.jpeg,.jbig2,.jb2,.PDF,.PNG,.JPG,.JPEG,.JBIG2,.JB2,.e
7 ps
8 Try typing <return> to proceed.
9 If that doesn't work, type X <return> to quit.
10

11 LaTeX Font Info: Trying to load font information for T1+txss on input
12 line 6
13 1.
14 (c:/texlive/2024/texmf-dist/tex/latex/txfonts/tltxss.fd
15 File: tltxss.fd 2000/12/15 v3.1
16)
17

18 LaTeX Font Info: Font shape `T1/txss/m/n' will be
19 (Font) scaled to size 17.09995pt on input line 61.
20

21 ! LaTeX Error: File `elsevier-logo-3p' not found.
22

23 See the LaTeX manual or LaTeX Companion for explanation.
24 Type H <return> for immediate help.
25 ...
26

27 1.61
28

29 I could not locate the file with any of these extensions:
30 .pdf,.png,.jpg,.mps,.jpeg,.jbig2,.jb2,.PDF,.PNG,.JPG,.JPEG,.JBIG2,.JB2,.e
31 ps
32 Try typing <return> to proceed.
33 If that doesn't work, type X <return> to quit.
34
35

36
37 ! LaTeX Error: File `SDlogo-3p' not found.
38

39 See the LaTeX manual or LaTeX Companion for explanation.
40 Type H <return> for immediate help.
41 ...
42

43 1.61
44

45 I could not locate the file with any of these extensions:
46 .pdf,.png,.jpg,.mps,.jpeg,.jbig2,.jb2,.PDF,.PNG,.JPG,.JPEG,.JBIG2,.JB2,.e
47 ps
48 Try typing <return> to proceed.
49 If that doesn't work, type X <return> to quit.
50
51

52 Underfull \hbox (badness 10000) has occurred while \output is active
53
54

55 []
56
57

58 Overfull \vbox (49.56999pt too high) has occurred while \output is active
59 []
60
61
62
63
64
65

1
2
3
4
5 Package hyperref Warning: Token not allowed in a PDF string (Unicode):
6 (hyperref) removing '\@ifnextchar' on input line 61.
7
8

9
10 Package hyperref Warning: Token not allowed in a PDF string (Unicode):
11 (hyperref) removing '\@ifnextchar' on input line 61.
12

13
14 Package hyperref Warning: Token not allowed in a PDF string (Unicode):
15 (hyperref) removing '\<def>-command' on input line 61.
16

17
18 Package hyperref Warning: Token not allowed in a PDF string (Unicode):
19 (hyperref) removing '\cnotenum' on input line 61.
20

21
22 Package hyperref Warning: Token not allowed in a PDF string (Unicode):
23 (hyperref) removing '\<def>-command' on input line 61.
24

25
26 Package hyperref Warning: Token not allowed in a PDF string (Unicode):
27 (hyperref) removing '\@corref' on input line 61.
28

29 [1{c:/texlive/2024/texmf-
30 var/fonts/map/pdftex/updmap/pdftex.map}{c:/texlive/202
31 4/texmf-dist/fonts/enc/dvips/base/8r.enc}
32
33]
34

35
36 Package natbib Warning: Citation `losing2022multi' on page 2 undefined on
37 input
38 line 64.
39

40
41 Package natbib Warning: Citation `losing2022newbee' on page 2 undefined
42 on input
43 line 64.
44

45
46 Package natbib Warning: Citation `xsens' on page 2 undefined on input
47 line 64.
48

49
50 Package natbib Warning: Citation `foot_pressure' on page 2 undefined on
51 input
52 line 64.
53

54
55 Package natbib Warning: Citation `tonsens2020high' on page 2 undefined on
56 input
57 line 64.
58
59
60
61
62
63
64
65

Package natbib Warning: Citation `losing2022multi' on page 2 undefined on input line 64.

Package natbib Warning: Citation `heading-direction' on page 2 undefined on input line 68.

Package natbib Warning: Citation `kiernan2023unsupervised' on page 2 undefined on input line 71.

Package natbib Warning: Citation `reid_imu_running_event_id' on page 2 undefined on input line 71.

Package natbib Warning: Citation `kuderle2024gaitmap' on page 2 undefined on input line 71.

Package natbib Warning: Citation `small2024self' on page 2 undefined on input line 71.

Package natbib Warning: Citation `small2022a' on page 2 undefined on input line 71.

Package natbib Warning: Citation `kiernan2023unsupervised' on page 2 undefined on input line 71.

Package natbib Warning: Citation `reid_imu_running_event_id' on page 2 undefined on input line 71.

Package natbib Warning: Citation `wu2002isb' on page 2 undefined on input line 74.

Package natbib Warning: Citation `kiernan2023unsupervised' on page 2 undefined on input line 74.

Package natbib Warning: Citation `rossler2024spatiotemporal' on page 2
undefined
on input line 75.

Package natbib Warning: Citation `parak2015evaluation' on page 2
undefined on i
nput line 78.

Package natbib Warning: Citation `rossler2024spatiotemporal' on page 2
undefined
on input line 78.

Package natbib Warning: Citation `benson2019automated' on page 2
undefined on i
nput line 81.

Package natbib Warning: Citation `auvinet2002runner' on page 2 undefined
on inp
ut line 81.

Package natbib Warning: Citation `greene2010adaptive' on page 2 undefined
on in
put line 81.

Package natbib Warning: Citation `mcgrath2012gyroscope' on page 2
undefined on
input line 81.

Package natbib Warning: Citation `fadillioğlu2020automated' on page 2
undefined
on input line 81.

[2]

LaTeX Warning: File `performance_surface.png' not found on input line 90.

! Package pdftex.def Error: File `performance_surface.png' not found:
using dra
ft setting.

See the pdftex.def package documentation for explanation.
Type H <return> for immediate help.

...

1.90 ... 0 0 2.5cm, clip]{performance_surface.png}

Try typing <return> to proceed.

If that doesn't work, type X <return> to quit.

LaTeX Font Info: Trying to load font information for T1+cmmt on input line 9

0.

(c:/texlive/2024/texmf-dist/tex/latex/base/t1cmmt.fd

File: t1cmmt.fd 2023/04/13 v2.5m Standard LaTeX font definitions

)

Overfull \hbox (29.7059pt too wide) in paragraph at lines 101--122

[][]

[]

Overfull \hbox (7.79991pt too wide) in paragraph at lines 134--148

[][]

[]

LaTeX Warning: `!h' float specifier changed to `!ht'.

Package natbib Warning: Citation `greene2010adaptive' on page 3 undefined on input line 161.

Package natbib Warning: Citation `mcgrath2012gyroscope' on page 3 undefined on input line 161.

Package natbib Warning: Citation `dumphart2025effect' on page 3 undefined on input line 161.

Package natbib Warning: Citation `panebianco2018analysis' on page 3 undefined on input line 164.

[3{c:/texlive/2024/texmf-dist/fonts/enc/dvips/cm-super/cm-super-t1.enc}]

Package natbib Warning: Citation `kiernan2023unsupervised' on page 4 undefined on input line 171.

Package natbib Warning: Citation `losing2022newbee' on page 4 undefined on input line 171.

No file main.bbl.

Package natbib Warning: There were undefined citations.

[4] (./main.aux)

LaTeX2e <2024-06-01> patch level 2
L3 programming layer <2024-05-27>

Package rerunfilecheck Info: File `main.out' has not changed.
(rerunfilecheck) Checksum:
95044EE5DD7A64D542DCA9EB5F791192;1397.
)

Here is how much of TeX's memory you used:

13769 strings out of 473583
213044 string characters out of 5732343
1951908 words of memory out of 5000000
36404 multiletter control sequences out of 15000+600000
641069 words of font info for 197 fonts, out of 8000000 for 9000
1141 hyphenation exceptions out of 8191
75i,15n,79p,110lb,440s stack positions out of
10000i,1000n,20000p,200000b,200000s
<c:/texlive/2024/texmf-dist/fonts/type1/public/txfonts/rtcxr.pfb><c:/texlive/2024/texmf-dist/fonts/type1/public/txfonts/rtxi.pfb><c:/texlive/2024/texmf-dist/fonts/type1/public/txfonts/rtxmi.pfb><c:/texlive/2024/texmf-dist/fonts/type1/public/txfonts/rtxr.pfb><c:/texlive/2024/texmf-dist/fonts/type1/public/cm-super/sfttl1000.pfb><c:/texlive/2024/texmf-dist/fonts/type1/public/txfonts/txsy.pfb><c:/texlive/2024/texmf-dist/fonts/type1/urw/helvetica/uhvr8a.pfb><c:/texlive/2024/texmf-dist/fonts/type1/urw/times/utmb8a.pfb><c:/texlive/2024/texmf-dist/fonts/type1/urw/times/utmr8a.pfb><c:/texlive/2024/texmf-dist/fonts/type1/urw/times/utmr8a.pfb>

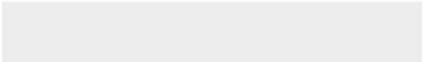
Output written on main.pdf (4 pages, 104847 bytes).

PDF statistics:

146 PDF objects out of 1000 (max. 8388607)
117 compressed objects within 2 object streams
22 named destinations out of 1000 (max. 500000)
97 words of extra memory for PDF output out of 10000 (max. 10000000)

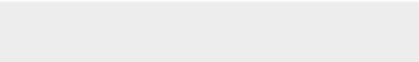
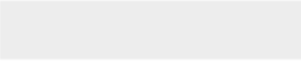


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elsarticle-num.bst



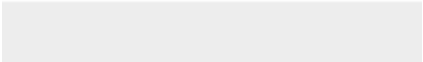


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LaTeX source files
elsarticle.cls



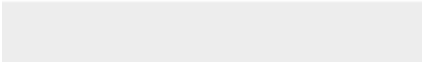


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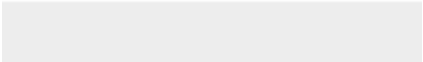


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elsevier-logo-3p.eps



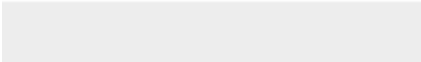


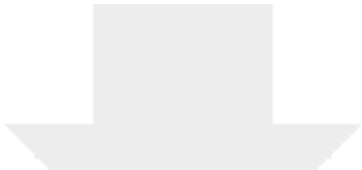
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LaTeX source files
elsevier-logo-3p.pdf





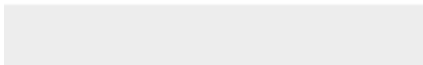
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LaTeX source files
performance_surface.png





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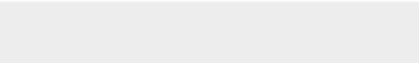
LaTeX source files
README-ecrc





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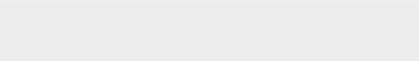
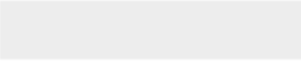
LaTeX source files
SDlogo-3p.eps





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LaTeX source files
SDlogo-3p.pdf



Declaration of interests

☐The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☒The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Philippe C. Dixon reports financial support was provided by Quebec Research Fund. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.
